CCCA Cell-State Analysis — Zhang et al. 2019, Hematologic

Follicular Lymphoma · 10x · 5,645 malignant cells (4 samples, 2 patients)

GO / NO-GO GATES

Gate 1 — mixing & continuity **GO**

Harmony(theta=10) integrates the clonal FL batch: same-sample neighbour fraction 0.975 -> 0.453, iLISI median 1.00 -> 2.26 of 4. One connected component (0 islands). theta=2 leaves a per-patient archipelago; judged on the regenerated embedding + per-neighbourhood iLISI, not a scalar: mixed AND continuous.

Gate 2 — module geography **GO**

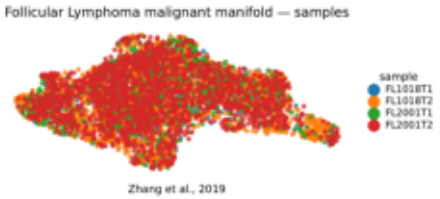
Non-branching case: a single (Gast/pSom)->Crest developmental gradient. Gast & pSom overlap at the origin (joint enrichment 1.00, smoothed r=-0.16), both oppose Crest (Gast<->Crest r=-0.55, pSom<->Crest r=-0.58). Along pseudotime Gast falls (rho -0.48), Crest rises to a terminal tip (decile pt=0.58). joint pSom&Crest enrichment 0.23 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — cross-repression **NA**

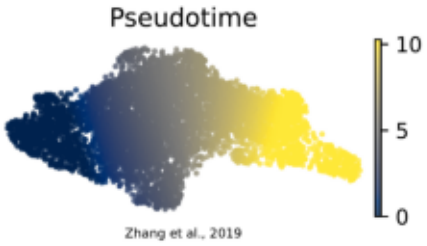
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced, per the deliverable spec).

STEP FIGURES

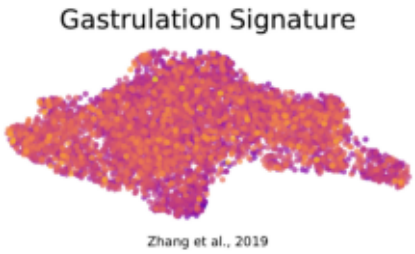
01 · Samples (Gate 1)



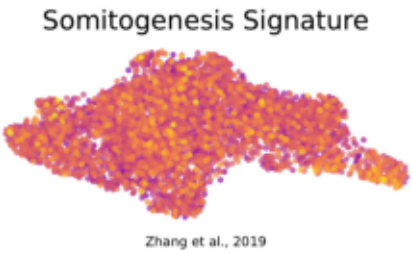
02 · Pseudotime



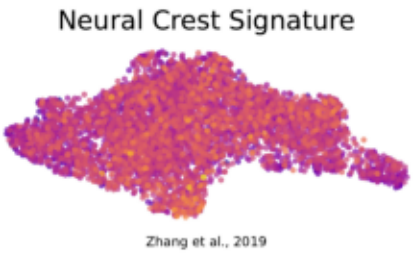
03 · Gastrulation



03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs



Dataset: Follicular lymphoma malignant B-cell compartment (Zhang 2019), 10x, 5,645 malignant cells over 4 samples / 2 patients (primary + progressed/transformed); single low-cycling body (~14% cycling).

Verdict: Pattern reproduces in the non-branching form (Linear-Gast): one continuous gastrulation -> neural-crest axis, Crest terminal, Gast globally strongest. G1 GO / G2 GO / G3 NA. EMT-TFs ZEB2 & ZEB1 (both terminus-loaded).

PARAMETER CHOICES

Sample threshold	>=100 (floor; largest gap ratio 2.74, no dominant cliff) -> 4/4 samples, 5,645 cells
Overdispersed genes	636 (scFates find_overdispersed)
Integration	Harmony on sample, theta=10 (mixing improves monotonically; clonal FL patient batch)
Embedding	n_pcs=20, n_neighbors=30, UMAP spread=1.3; 0 islands removed
Geometry	Non-branching -> scFates principal curve, Nodes=20, epg_lambda=0.3, epg_mu=100
Topology check	2 tips / 0 forks; fold self-approach 2.96 (fold-free); pt vs cycling rho +0.29
Root	tip 2 = Gast/pSom-high, Crest-low origin (index may shift a few between runs)
Module means (raw)	Gast +0.012 / pSom +0.006 / Crest +0.009 (Gast strongest -> Linear-Gast)
EMT-TFs shown	ZEB2 & ZEB1 only (>3% detected; both terminus-loaded, ZEB2 earlier; no root-loaded TF)